

Exploring the Opportunities and Challenges of Enabling Clinical-friendly Drug Psychotherapy with Virtual Reality and Biofeedback Technologies

Min-Wei Hung
National Tsing Hua University
Hsinchu, Taiwan
mwhung@mx.nthu.edu.tw

Chi-Ting Hou
National Tsing Hua University
Hsinchu, Taiwan
dafdafdaf0411@gapp.nthu.edu.tw

Chieh-Jui Ho
National Tsing Hua University
Hsinchu, Taiwan
jerryjp3914@gapp.nthu.edu.tw

Chien Wen (Tina) Yuan
National Taiwan Normal University
Taipei, Taiwan
tinayuan@ntnu.edu.tw

Nanyi Bi
National Taiwan University
Taipei, Taiwan
nanyi.bi@gmail.com

Shu-Huei Chen
Taipei City Hospital Songde Branch
Taipei, Taiwan
T0757@tpech.gov.tw

Ming-Chyi Huang
Taipei City Hospital Songde Branch
Taipei, Taiwan
mch@tpech.gov.tw

Chuang-Wen You
National Tsing Hua University
Hsinchu, Taiwan
cwyou@mx.nthu.edu.tw

Abstract

Drug addiction is a chronic disorder associated with many of the emotional or cognitive problems observed in addicted individuals. As a result, drug-addicted patients often display deficits in their mind-body awareness and easily ignore or fail to identify emotional or environmental cues. This study conducted a needs assessment study to gain guidelines for incorporating virtual reality and biofeedback technologies to assist psychotherapists in raising patients' awareness and identifying their cues in psychotherapy. Our results identify current difficulties to reinforce mind-body awareness and correct attributions of craving stimuli. Also, we summarized concerns and the potential to promote the effectiveness of psychotherapy when applying virtual reality and biofeedback technologies to induce cravings. We also proposed a preliminary design of technology solutions to incorporate clinical-friendly VR-based craving inducing along with real-time biofeedback for enabling psychotherapists to jointly review the induced craving events based on the collected information.

CCS Concepts

• **Human-centered computing** → **Empirical studies in ubiquitous and mobile computing**.

Keywords

Virtual reality, biofeedback, psychotherapy, drug addiction

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1 Introduction

Drug abuse is one of the common addictions that damage the central nervous system and present various symptoms, including emotional irritability, depression, anxiety, impairment of attention or memory, or even hallucinations and delusions, thereby affecting an individual's function in daily life, work and relationship. Among those treated, 40-60% of patients experience relapses after leaving rehabilitation centers [9]. Cognitive behavioral therapy (CBT) [32] is one of the approaches to avoid relapses and effectively assist the addiction patients in the process of recovery, in which the role of Virtual Reality (VR) has been explored in the past decades. In CBT for drug addiction, the Relapse Prevention (RP) model [26] serves as a guideline for therapists to provide specific intervention strategies according to the stage of relapse process during treatment. One of the many challenges at the early stage of treatment is that patients often display a lack of ability to identify the factors that induce their craving for drugs and fail to avoid them. Without such an awareness, they would act on their impulse and follow their habits of decision-making to resolve their craving, leading to a series of relapse behaviors (i.e., buying or using drugs). In response to this challenge, therapists would focus on guiding patients to recollect the moments when craving arose, and review the triggering factors, bodily responses and mental experiences under the circumstances. Only when patients have the capacity to identify triggering factors and improve their awareness of craving

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can they practice self-regulation and coping skills, thereby blocks the inertial relapse process at the right time.

Existing research regarding VR applications in psychotherapy has been centered around PTSD [24], anxiety [27, 37, 47], depression [12, 27], as well as addiction [14, 23, 41]. Virtual Reality (VR) has been introduced in CBT as a tool for assessment and intervention, and has demonstrated its potential in VR exposure therapy (VRET) [39]. However, in treating drug addiction, the focus is mostly on the effectiveness and validity to induce craving [14]. Moreover, integration of VR and biofeedback has been explored specifically in the application domain of relaxation and gaming experience. To the best of our knowledge, there is an absence of research on the qualitative insights on the design space of VR experience combined with biofeedback in clinical settings for drug addiction that aligned with treatment goals. We aim to start from the users' perspective, and integrate therapists' professional opinions and the counseling experience with patients dealing with drug abuse, in an effort to further develop a treatment-oriented system that promotes therapy quality.

In this study, we interviewed five experienced psychotherapists in treatment for drug addiction and biofeedback for needs assessment, and used thematic analysis to identify difficulties in psychotherapy to promote patient's awareness of craving and issues regarding the application of VR and biofeedback in drug addiction. Based on these insights, we propose a preliminary system design that provides a clinical-friendly VR experience incorporating timely biofeedback, supporting psychotherapists and patients to effectively reflect on craving events with the hope to facilitate treatment progress of drug addiction.

2 Background & Related Work

Relapse prevention (RP) model [26] in CBT emphasizes that drug addiction is a very complex "chronic disease", involving physical, psychological, and social problems, which patients have to live with throughout their lives. Patients may constantly face the risk of relapses triggered by various extrinsic or intrinsic factors during abstinence. Therefore, the RP model proposes that one of the objectives in the initial treatment stage is to increase the awareness of craving sensation and identify correct risk factors that induce such impulse. Otherwise, exacerbated by impaired judgment, patients will fall back into old habits with fixed thinking and behavioral patterns when encountered by drug-associated factors that lead to a series of impulsive behaviors. Only with sufficient ability to recognize and observe these experiences, patients can start practicing cognitive regulation and behavioral strategies to change automated thoughts or irrational beliefs [26, 31]. To understand how VR combined with biofeedback can be integrated into CBT to promote drug addiction treatment, we summarized studies on how to apply (1) virtual reality and (2) biofeedback technologies to assist psychotherapists in addiction treatment, described as follows.

2.1 Applying Virtual Reality Technology to Assist Psychotherapists

The incorporation of Virtual Reality (VR) in CBT has been the interest of research in the field of psychotherapy to examine the

effect of virtual cues and compare them with in vivo cues in exposure therapy. In traditional cognitive behavioral therapy, exposure techniques are often used for treating anxiety, fear, trauma, or addiction [27], by exposing the patient to charging stimuli in a safe psychological and physical environment, therapists can challenge their maladaptive cognition, or encourage them to practice coping skills. Due to technical limitations, exposure therapy usually uses in vivo cues to help the client re-live in a mental context that induces responses, such as craving in addiction treatment. The application of VR in psychotherapy may improve the quality of cues by making vivid visual reconstruction, providing a first-person immersive experience in a virtual scene, and adding details that are difficult to reproduce in reality. With the assistance of VR, psychotherapists can use lifelike stimuli to assess craving, and improve the effectiveness of exposure techniques (Virtual Reality Exposure Therapy, VRET). This application allows the patients to gain insight into the internal automation process or practice the skills of rejecting triggers under the supervision and guidance of the psychotherapist, and hopefully transfer these skills to real-life situations [14]. Besides VRET, VR application in mental health has been done around topics including mindfulness, relaxation [30], stress management [15], self-injury [25], depression [27], and anxiety [37]. Despite the many advantages of VR applications, there are still limitations for the therapeutic use of VR. For instance, pre-designed content without customization that is far from the user's real-life experience may result in difficulties for users to immerse themselves in the virtual environment and transfer improvements in treatment into daily life [21]. So far, studies have supported the potential of VR psychotherapy of drug addiction [14, 23, 41], and emphasized the importance of involving relevant stakeholders in the design process for VRET [7, 19]. However, how to integrate VR technology into the clinical practice of drug addiction treatment and align with the treatment goal still invites further exploration from a user-centered perspective.

2.2 Combining Virtual Reality with Biofeedback Technology to Assist Psychotherapists

Biofeedback has been used in psychotherapy for decades, as it enables the psychotherapist to bring the physical state of the patient into the discussion of cognitive and emotional issues. Biofeedback therapy [20] is the use of monitoring devices to visualize physiological responses (including changes in skin conductance level, heart rate, and respiration rate), allowing not only the patient to understand the interaction between body and mind but also psychotherapists to gain an objective understanding of patient's state. Therapists can then use collected data to guide the patient to improve mind-body awareness and learn to self-regulate along with visual representations [16–18, 34]. When assessing craving with biofeedback in drug addiction psychotherapy, the therapists help attach biofeedback sensors to patients and record the baseline of their physiological state. Afterward, therapists instruct patients to recall the situation in which their drug craving occurred and collect patients' bodily responses as indicators of craving. Once the patient's state is aroused, it is essential for the therapists to

make sure that the aroused state of patients goes back to baseline by providing self-regulation techniques [6, 17, 46].

In recent years, the combination of VR and biofeedback has become popular in mindfulness and relaxation exercises by providing users with relaxing multi-sensory cues (e.g., simulating natural environment) accompanied with visual biofeedback (such as breathing and heartbeat), so that users can see the physiological changes produced by practice self-regulation in real-time [30, 33, 38, 42]. In addition, VR psychotherapy often incorporates biofeedback to monitor patient's physiological responses to a target virtual situation, so that he/she can review the readings afterwards to pinpoint and reflect on specific, meaningful time points (e.g., when their heart rate spikes) in the VR experience [27]. Repetto et.al [37] presented a clinical protocol for treating anxiety based on a biofeedback enhanced-VR system, and other VR-based approaches combined with biofeedback are explored in [12, 33, 47] to address issues around depression or anxiety. This combination method resembles traditional biofeedback therapy, but also gives rise to issues that have not been discussed previously. For example, the integration with biofeedback lacks immediacy and patients can not access readings in real time. Without biofeedback in the VR experience, they cannot see the changes in their physiological responses in time, thus the effect of enhancing self-awareness may be limited.

Based on the literature review, there is strong potential for VR and biofeedback applications to thrive in promoting various issues in mental health. However, so far little has been discussed for drug addiction and the design approach that is clinical-friendly. In addition, researchers have brought up concerns and suggestions for future VR applications in psychotherapy. Besides the general attempt to lower the financial cost, some are concerned about lack of resources for therapists to receive VR training [8] and transferring improvements from treatment to the real world [28], indicating the unique requirements for therapists to incorporate VR in treatment, as well as needs to optimize the design of VR experience that bridges the gap between current practice and VR application. In light of this need, this study seeks to explore this question through interviews and questionnaires on how the immersive experience provided by VR can incorporate biofeedback in drug addiction treatment, as well as engage relevant stakeholders to identify system design considerations for psychotherapists and drug addiction patients.

3 The Needs Assessment Study

Needs Assessment was based on interviews with psychotherapists. From Taipei City Psychiatrist Center, we recruited 5 psychotherapists (2 male and 3 females; PT01 ~ PT05) aged from 28 to 46 years old ($M = 34$ years old; $SD = 7.38$), with 1.5 ~ 16 years of experience in conducting psychotherapy in the psychiatric center. All participants have at least a passing knowledge about what virtual reality (VR) and biofeedback are and have heard of examples [15, 25, 27, 30, 37] of applying these technologies in psychotherapy. Moreover, each psychotherapist has either experience in offering biofeedback training for stress or anxiety management or conducting psychotherapy on drug-addicted patients for promoting their motivation or improving the abilities of recognizing cues in daily life.

We first conducted semi-structured interviews with psychotherapists to explore the opportunities and difficulties of applying VR and biofeedback technologies to assist them in conducting psychotherapy of drug addiction. Interview questions included: (1) understanding current practices of psychotherapy on drug-addicted patients, (2) identifying when and how to apply biofeedback during sessions of psychotherapy, and (3) exploring opinions and concerns towards using VR and biofeedback to improve mind-body awareness [36, 43] and attributions of craving stimuli during psychotherapy. Our objective was to characterize the counseling interactions between psychotherapists and drug-addicted patients in psychotherapy and collect information pertaining to introducing these VR and biofeedback technologies in psychotherapy to promote the effectiveness of sessions. Each participant was reimbursed with NTD 250 (~USD 16.5). We audio-recorded participants' responses towards these interview questions, which were transcribed to identify major themes based on thematic analysis [11] described as follows.

3.1 The Difficulties of Reinforcing Mind-body Awareness and Correct Attributions of Craving Stimuli

Drug addiction is a chronic and relapsing disorder. With repeated use, drugs can cause long-term changes in the brain systems. Studies in individuals who used drugs [13, 29, 44] show that severe structural and functional changes in brain regions associated with emotion or memory, which caused many of the emotional or cognitive problems observed in these individuals. In addition, individuals with problems of substance abuse tend to suppress or avoid mental and physical experiences in maladaptive manners, such as using drugs to alleviate pain, craving the next high to escape from a negative state [48]. As a result, drug-addicted users often display deficits in their mind-body awareness; that is, they are insensitive to or unaware of the interaction dynamics between inner bodily sensation and mental state [36, 43]. Thus, they often have little understanding of what craving feels like and how to cope with it in adaptive ways [40, 45, 48]. As commented by PT01, "*when I discussed with a patient about his mind-body awareness towards craving, the patient would recall that he had no special feeling in the morning of that day because he did not perceive that he was craving for drugs. He would feel or would not think that he was craving ... but actually, he was and his body had some reactions due to craving. I think this is not a problem caused by the patient's recall ability but due to his problematic awareness ability*".

Furthermore, due to their overlook of craving sensations, they would easily ignore or fail to identify what emotional or environmental cues have caused the craving for the drugs or attribute such causes to incorrect life events or environmental factors. Therefore, all participants commented on how their patients would react during clinical sessions. For example, PT03 mentioned "*There are many patients who thought that 'My craving was triggered because I saw the drug. It has nothing to do with my moods ... or with passing the location where I bought the drugs.' They would not feel in that way. So, they probably did not know, for example, these external or internal cues were responsible for triggering their drug use. He would think that 'I used drugs because I was craving for them. That's all.' He probably did not know the seriousness of those cues to drug uses*".

To reinforce the mind-body awareness and correct attribution of craving stimuli, psychotherapists have to expose stimuli to induce cravings and discuss with patients to sort out what stimuli would induce their cravings for drugs and how to cope with these identified stimuli in real life. However, in current practices, participated psychotherapists responded that they can only use limited resources, e.g., a paper-based tool or videos, to expose cues to patients, which would be less effective than other vivid and interactive media, e.g., virtual reality video, that can facilitate patients to imagine real-life experience analogous to the VR experience. For example, PT05 mentioned *“It would be better if the video includes scenes exactly the same as those appearing in patients’ real life. I think it would be best for exposing cues to patients. However, I know there are limitations ... But it is just like watching a movie. Although the movie is not about our own stories, somehow ... we would be affected by the scenes or empathize with the characters in the movie. [...] On the other hand, if I just let a patient see a picture or verbally describe a story, patients would not be fully engaged, making it harder to show empathy with the content and relate the scenario to their personal life”*.

To introduce new interactive media to expose cues in a clinical-friendly way, we have to consider the structure of current sessions. Participated psychotherapists responded that a session of psychotherapy typically lasts 45 ~ 50 minutes and usually consists of two parts, which are cue-exposing and follow-up intervention. Bringing in new interactive media can help patients imagine but might also introduce other factors (e.g., length or numbers of cues in the VR video) impacting the treatment effectiveness when providing interventions, which were expressed by most psychotherapists. For example, PT04 mentioned *“The advantage of exposing cues with VR videos is that it explicitly puts the patients in a virtual scenario. However, if researchers embed too many cues in the video in different scenes, it is hard for psychotherapists to identify the right cues corresponding to these craving reactions. If I deliver the intervention afterwards, I would experience difficulty to identify the most accurate cue for a craving or properly stop patients’ craving.”*.

3.2 Concerns Associated with Applying Virtual Reality Technology to Induce Cravings

During stages of psychotherapy, psychotherapists would have different treatment tasks and deliver different tactical interventions to teach patients how to cope with cravings. Psychotherapists would assess patients’ progress and incorporate suitable tools to assist them in achieving treatment tasks. For example, PT03 expressed that she had concerns about the timing to apply Virtual Reality Technology to induce cravings. She commented *“[Would you have any concerns about applying Virtual Reality Technology to induce patients’ cravings?] Uh ... I would prefer to use this tool in the middle or later stages of psychotherapy. That is, after patients have learned some skills to cope with cravings, they would know what to do if their cravings were induced. They would feel safer with more resources at hand ... Then, I will consider to use this VR tool, starting from the lightest level of cues embedded in VR videos, to induce their craving.”*.

Using VR videos to expose cues helps patients to perceive and understand the content. However, most of the participated psychotherapists worried that the vivid and imaginary cues appearing

in the VR videos would directly render patients in high-risk situations (e.g., drug-using actions) and induce cravings that are too strong to be properly addressed in the intervention part. For example, PT03 raised her concern about the ethical issue of using this VR tool and commented *“Regarding exposing cues to patients through watching a VR video, I am concerned about the ethical issue ... because patients’ cravings for drugs are triggered ... [What if psychotherapists provide interventions to assist patients in handling the craving shortly afterwards?] I am worried that this might not be enough for patients who are just entering the initial phase of psychotherapy. That is, I am afraid that they will be triggered to buy drugs after the session ... ”*.

Moreover, experiencing virtual reality content requires patients to immersively enter the virtual space, which prevents psychotherapists who are in the physical space from directly interacting with patients. Since psychotherapists do not enter the virtual space, they need the corresponding information (e.g., what patients watch or how patients react) collected at the time instance when patients were induced to be craving. As PT02 commented *“If I don’t have the information about which part of videos patients currently watch when they respond with craving reactions, it would take more time to find out that relevant information afterwards. I would have to ask them in the following intervention phase about what they saw during the VR experience”*.

3.3 Using Biofeedback to Promote the Effectiveness of Psychotherapy

Traditionally, psychotherapists rely on behavioral observations and patients’ self-reports to identify when patients’ cravings are cued. However, given the individual differences, patients might react differently when craving. Sometimes, psychotherapists experience difficulty to identify if patients are craving without any objective indicators. Although psychotherapists can objectively identify when patients are induced to be craving through incorporating readings from biofeedback devices, they might also bring in other false detections of craving. A better way to review potential craving events from biofeedback calls for further research. Psychotherapists can quickly go through all potential time instances when patients react differently (e.g., higher heart rate variability or skin conductance) to identify when patients are really craving for drugs to provide future intervention. For example, PT05 commented *“When I assessed the stressed event recalled by patients [in biofeedback training for stress management], I would need to know what patients were thinking. [...] I did not go through every stressed event. Because there are lots of glitches in biofeedback readings in 10-min training, I would briefly ask patients if they can concentrate on the breathing practice or be just distracted by other things. [...] Moreover, I would closely observe patients’ reactions, as they might accidentally grab the wires or yawn and their breath would be significantly impacted, which results in a drastic change in biofeedback readings.”*.

Given the impact of drugs on patients’ cognitive abilities, psychotherapists suggested that this biofeedback information should be used to simultaneously remind patients during virtual reality experience. For example, PT05 commented *“I think it is better to give real-time biofeedback when patients watch the VR video. Since their function of short-term memory has likely been damaged [due to*



Figure 1: Screenshot of four key cues appearing in the virtual reality video which was created based on free assets from the Unity Asset Store [4], Sketchfab [3], and Mixamo [1].

long-term drug uses], it would be better to provide real-time biofeedback while watching the VR video. That is, they will perceive this reminder when their physiological readings go higher, and they would have a clear impression of the changes that just happened in their biofeedback readings and perhaps start connecting the dots. On the contrary, if you just give patients data logs of those readings and discuss with them afterwards regarding those significantly-changing events, some patients may make excuses for themselves. With real-time biofeedback, we might be able to reduce the chances that patients insist on their biased opinions”.

4 Preliminary System Design for Promoting the Effectiveness of Psychotherapy

Based on the salient themes extracted from the previous needs assessment interview study, we proposed our preliminary system design, which consists of (1) clinical-friendly VR-based craving inducing, (2) real-time craving reminding, and (3) correlation visualization.

4.1 Clinical-friendly VR-based Craving Inducing

According to the Relapse Prevention model [26], psychotherapists would teach patients how to identify and cope with craving stimuli to prevent high-risk situations. Instead of directly triggering patients craving with virtual reality videos emulating with high-risk

situations, e.g., drug-using scenarios [14], we step back to visualize craving stimuli. After consulting with psychotherapists, we picked four types of craving stimuli (i.e., cues shown in Figure 1) and embedded them in our preliminary version of virtual reality video, which is described as follows.

Figure 1 demonstrates four screenshots extracted from the current version of the virtual reality video. The video first shows a bedroom imagined as a room at their home. While patients wear the VR helmet (e.g., HTC Vive headset [5]) and watch the video, they would see a drug-free calendar showing that they are free from drugs for about half of a month. Then, patients would receive a phone invitation from their friend (i.e., Joe) to his party (the first cue in Figure 1(a)). After a while, the scene showing in the video would transition to another Joe’s house emulating that they are attending the party. Patients would see a bunch of people drinking alcohol and having fun together. On the table, there are food, bottles, and a drinking cup with a straw (resembling a home-made plastic bong in Figure 1(b)). Then, patients would walk to the toilet, where they would receive a text message from a drug dealer (the third cue in Figure 1(c)). Finally, patients would enter another bedroom where people are excitedly dancing and get an after-party invitation from a friend (a cue reminding them using drugs in the after-party in Figure 1(d)). Note that the design of cues in the VR content needs to incorporate opinions from both psychotherapists and patients. We will continue to collect data from related stakeholders to improve the design of cues in the future.

4.2 Real-time Craving Reminding inferred from Biofeedback Readings

Biofeedback is a useful tool to demonstrate to patients the mind (i.e., cognitive or emotional reasoning) to body (i.e., physiological response) interaction [35]. Therefore, we use biofeedback devices, e.g., ProComp Infiniti System [2], to collect patients' physiological signals (e.g., heart rate variability, skin conductance, or skin temperature) in real time as the baseline and craving examples when exposed to drug-related and neutral VR video cues in the initial stage of psychotherapy. In the following sessions, we can predict if patients have elevated craving and physiological reactions after exposure to VR videos based on the machine learning model learnt from collected data [10, 22].

Once we can detect real-time craving based on the predictors, we can send real-time craving reminders to patients while watching a VR video to reinforce the link between physical reactions (i.e., body) and their cognitive process (i.e., mind) [35], and also to psychotherapists to pay more attention to patients' reactions. However, we have to carefully design the way to feedback the reminders without impacting the immersive video experience. One potential design direction is to feedback the reminders through other sensing modalities based on the type of data, e.g., tactile feedback for the heartbeat, to avoid distracting patients with visual indicators.

4.3 Correlation Visualization for Reviewing Craving Events

To facilitate psychotherapists in their discussion with patients in the intervention part of a session, a correlation visualization interface is required. The interface visualizes the segment of VR video, biofeedback readings, and psychotherapist's observation towards patients' conditions corresponding to a craving event. With these types of information (i.e., VR video, biofeedback readings, psychotherapist's observations), psychotherapists can first compare the biofeedback readings collected while watching the VR video with the baseline counterparts and also their observations towards patients' reactions to objectively filter out those false ones, e.g., drastically changes caused by posture changes. Then psychotherapists can jointly use these types of information to pinpoint the potential ones to discuss with patients to improve the mind-body awareness and the attribution of craving stimuli.

5 Conclusion & Future Work

In the future, we will interview more psychotherapists and drug-addicted patients to refine the design of this system. We will also prototype our system and further validate its potential and feasibility through conducting a field study involving psychotherapists and drug-addicted patients.

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